AMERICAN UNIVERSITY OF BEIRUT
MAROUN SEMAAN FACULTY OF ENGINEERING & ARCHITECTURE

Department of Industrial Engineering and Management (IEM)

Annual Worth Analysis

Engineering Economy

Annual Worth

An advantageous way of comparing alternatives



An Advantageous Way Of Comparing Alternatives

- AW ← Only one lifecycle of an alternative

- **Repeatability Assumptions**
  - **Equal-Service Requirement**
  - **Identical Replacement**
  - **Constant Cash Flows:** Costs and revenues won't change due to inflation in future cycles.

- No need for LCM or study period

AW – Example

Alt.	Description	Costs	Revenue	Expected Life	Salvage Value
A	A new line with 240,000 throughput and 98% eff	6 millions USD	5,000 USD per day	6	$ 500,000
B	A used line with 200,000 throughput and 96% eff	2.5 millions USD	2,000 USD per day	3	$ 200,000
C	A *co-developed* line with 240,000 throughput and 97% eff	5 millions USD	3,000 USD per day	4	$ 400,000

$$AW = -P\left(A/_{P}, i, n\right) + S\left(A/_{F}, i, n\right) + R_{year}$$

AW – Example

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AW - Example

- Ch5 example
- Alternative 1 is the best
- Required cashflow modifications
  - LCM
  - Study Period

Alternative	Initial Cost	Annual Revenue	Salvage Value	Expected Life
A	$ (6,000,000)	$ 1,825,000	$ 500,000	6
B	$ (2,500,000)	$ 730,000	$ 200,000	3
C	$ (5,000,000)	$ 1,095,000	$ 400,000	4

MARR	10%
LCM	12

Year	Alt. A	Alt. B	Alt. C
0	$ (6,000,000)	$ (2,500,000)	$ (5,000,000)
1	$ 1,825,000	$ 730,000	$ 1,095,000
2	$ 1,825,000	$ 730,000	$ 1,095,000
3	$ 1,825,000	$ (1,570,000)	$ 1,095,000
4	$ 1,825,000	$ 730,000	$ (3,505,000)
5	$ 1,825,000	$ 730,000	$ 1,095,000
6	$ (3,675,000)	$ (1,570,000)	$ 1,095,000
7	$ 1,825,000	$ 730,000	$ 1,095,000
8	$ 1,825,000	$ 730,000	$ (3,505,000)
9	$ 1,825,000	$ (1,570,000)	$ 1,095,000
10	$ 1,825,000	$ 730,000	$ 1,095,000
11	$ 1,825,000	$ 730,000	$ 1,095,000
12	$ 2,325,000	$ 930,000	$ 1,495,000
PW(i=10%)	$3,489,696	($1,464,017)	($2,699,351)

AW – Example

Alt.	Costs	Revenue	Expected Life	Salvage Value	AW
A	6 millions USD	5,000 USD per day	6	$ 500,000	$ 512,159
B	2.5 millions USD	2,000 USD per day	3	$ 200,000	$ -214,864
C	5 millions USD	3,000 USD per day	4	$ 400,000	$ -396,165

$$AW = -P\left(A/_{P}, i, n\right) + S\left(A/_{F}, i, n\right) + R_{year}$$

AW – Example

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$$AW = \underbrace{-P\left(A/_{P}, i, n\right) + S\left(A/_{F}, i, n\right)}_{\text{Capital Recovery } (CR)} + R_{year}$$

AW – Example

Alt.	Costs	Revenue	Expected Life	Salvage Value	AW	CR
A	6 millions USD	5,000 USD per day	6	$ 500,000	$ 512,159	$ -1.3 mil
B	2.5 millions USD	2,000 USD per day	3	$ 200,000	$ -214,864	$ -0.95 mil
C	5 millions USD	3,000 USD per day	4	$ 400,000	$ -396,165	$ -1.49 mil

$$AW = \underbrace{-P\left(A/_{P}, i, n\right) + S\left(A/_{F}, i, n\right)} + R_{year}$$

Capital Recovery (CR)

Amount you need to earn each year to recover investment + some rate of return (MARR) while considering salvage value

What is the AW Actually Telling Us?

$AW < 0$ 

- **Revenue Alternative**:
  - Fails to cover CAPEX, OPEX
- **Cost Alternative**:
  - Pure annual cost to run the system
  - E.g.: infrastructure, maintenance, …

$AW > 0$ 

- The investment is:
  - Recovering it CAPEX
  - Recovering its OPEX
  - Paying the required ROR (MARR)
  - Generating profit

AW for Independent Projects

Alt.	Description	Costs	Expected Life	Revenue
1	Install an online inspection system.	100,000 Initial + 2,000 per year	10	15,000 USD per year
2	Replace existing guiding systems with more accurate ones.	± 16,000 Initial	5	5,000 USD per year
3	Buy a new Cutting die made of tungsten carbide material.	300,000 USD	2	40,000 USD per year

Select All alternatives with $AW \geq 0$

Permanent Investments

Not all projects have a limited lifetime

Recall

Chapter 4

CC Analysis - Procedure

1. Cash flow diagram over **at least two cycles** of periodic cash flows
2. One time cash flows → get the PW
3. Recurring cash flows with a period of n_R (e.g., $\$R$ every n_R years):
   1. Find equivalent uniform annual worth:

$$A_R = R(A/F, i, n_R)$$

   2. Find equivalent CC for the A_R series:

$$CC_R = \frac{A_R}{i}$$

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Perpetuity – Infinite equivalent annual cash flows



CC

$$AW = CC \times i$$



AW

What it is?

The Present Worth (PW) of a project with an infinite life.

The equivalent **Annual Worth (AW)** of those infinite cash flows.

It answers:

"How much money do we need to put in the bank TODAY to fund this project forever?"

"What is the equivalent budget we need to spend/earn EVERY YEAR, forever?"

Perpetuity – Infinite equivalent annual cash flows



CC

$$AW = CC \times i$$



AW

The city is building a permanent suspension bridge. The MARR is **5%**.

- **Initial Construction:** $50,000,000
- **Routine Maintenance:** $1,500,000 / year
- **Major Deck Resurfacing:** $10,000,000 every 15 years

The City Council
Fund the project by issuing Bond

The Private Partner
BOT over 30 years → Set up a toll

AW of Infinite Projects - Example

A government agency is planning a permanent hydroelectric and irrigation dam with a MARR of 6% per year.

- Construction: 10m
- AOC: 200k
- Structural inspection at Year 5: 150k
- Overhauling every 12 years: 500k

What is the AW (perpetuity) of this project? And what does it mean?